

Motor Insurance Pure Premium

Prediction

An Interpretable Neural Additive Model Based on Multi-Task Learning

**Undergraduate
Thesis Defense**

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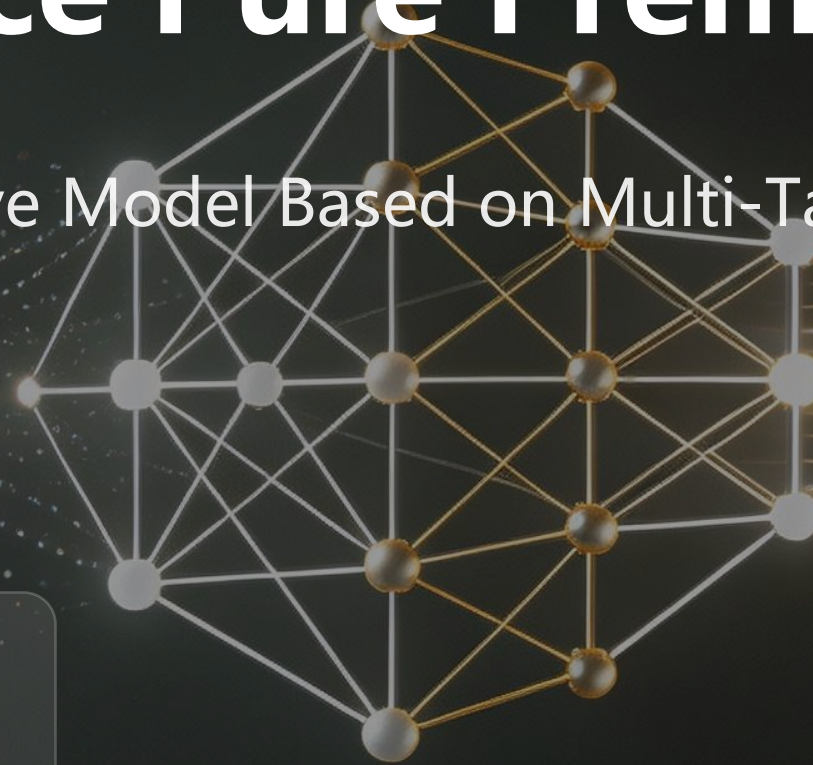
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Summary of key findings, contributions, limitations, and future research directions

An iceberg floating in a grey sea. The tip of the iceberg, which is above the water line, is constructed from a stack of light grey and tan cubes. The much larger part of the iceberg, which is submerged below the water line, is constructed from a stack of dark grey and black cubes. A large white number '01' is centered over the water line, with the '0' partially on the submerged part and the '1' on the visible tip.

01

Research Background and Significance

RESEARCH BACKGROUND AND SIGNIFICANCE

Industry Background: Scale and Challenges



Large Market Scale

Motor insurance premiums exceeded RMB 913 billion in 2024, accounting for over 54% of total non-life premiums; vehicle ownership reached 353 million, and the market continues to expand.



Underwriting Profitability Challenge

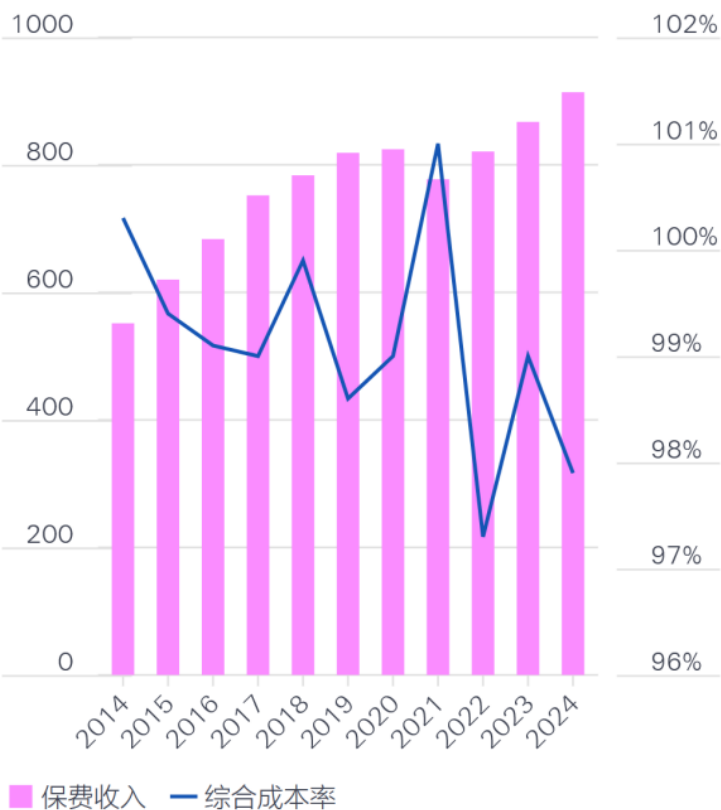
The industry combined ratio has long fluctuated around break-even, leaving underwriting profitability fragile. Accurate pricing is a key lever for improving underwriting margins.



Core Industry Challenge

How can data and analytics enable sound, granular, and accurate pricing in an intensely competitive market?

车险保费收入（单位：十亿元人民币）
和综合成本率



Problem Statement: The Accuracy–Interpretability Trade-off

Traditional Actuarial Models (e.g., GLM)



Strength: transparent structure and strong interpretability, well suited to regulatory and audit requirements.



Limitation: linear assumptions restrict complex nonlinear risk relationships and can limit predictive accuracy.

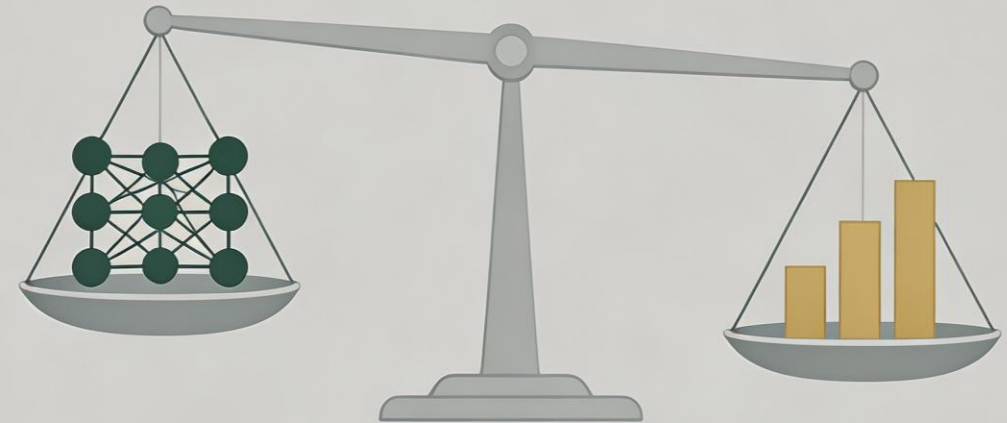
Modern Machine-Learning Models (e.g., GBM)



Strength: powerful nonlinear modeling capacity that can extract complex structure and improve risk prediction.



Limitation: black-box decision logic is difficult to interpret directly and harder to audit.



Core Tension: Trade-off and Balance

How can we move beyond the apparent conflict between predictive accuracy and interpretability? This is the central question of the study.

Research Objectives and Significance

Research Objectives



01. Build a Multi-Task Neural Additive Model (MTL-NAM)

Jointly predict claim frequency and severity in a unified framework and leverage shared information across tasks.



02. Balance Predictive Accuracy and Interpretability

Improve pure premium prediction while preserving inherent interpretability rather than relying on a black box.



03. Systematic Evaluation on Real-World Data

Use the freMTPL motor insurance dataset for controlled benchmark experiments evaluating both effectiveness and interpretability.

Research Significance



Methodological Contribution

Explore the integration of multi-task learning and structurally interpretable models in actuarial pricing, offering a new perspective on the performance–interpretability trade-off.



Practical Value for Insurance Risk Management

Provide a reproducible modeling approach that supports more accurate and transparent pricing and more granular risk management.

02

Research Design and Model Development

RESEARCH DESIGN AND MODEL CONSTRUCTION

Research Framework and Technical Workflow



01 / Problem Identification

Identify the central tension between predictive accuracy and interpretability in motor insurance pricing.



02 / Method Development

Apply MTL-NAM to pure premium modeling and jointly learn claim frequency and severity.



03 / Empirical Validation

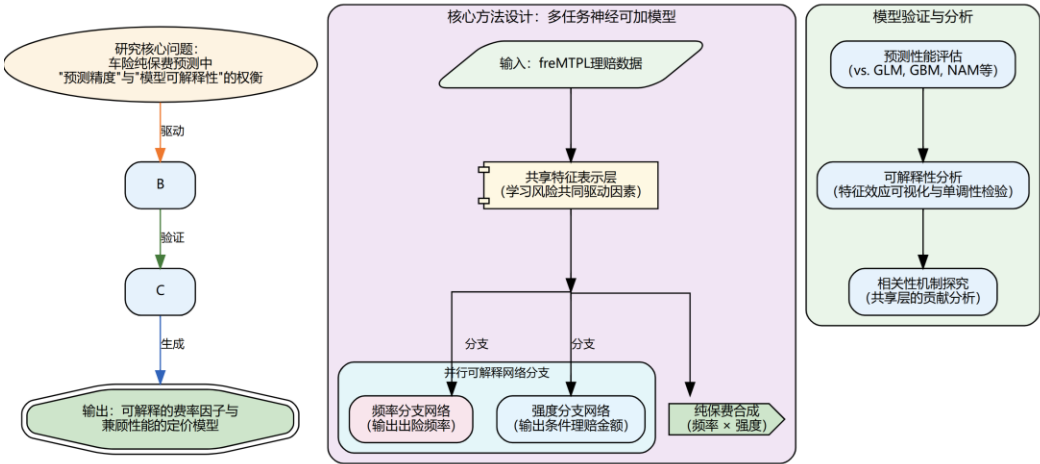
Using freMTPL and six-fold cross-validation, compare MTL-NAM out of sample with GLM, GBM, and NAM using NLL, RMSE, and MAE, together with shape functions and SHAP for interpretability.



04 / Implications

Assess whether MTL-NAM can combine strong predictive performance with interpretability and provide practical evidence for granular motor insurance pricing.

Research Logic



A closed-loop workflow: problem → method → validation → application

freMTPL French Motor Third-Party Liability Dataset

Variable	Type / Values		Description
IDpol	Integer ID		Policy ID
ClaimNb	Non-negative integer		Claims during observation period
Exposure	Continuous in [0, 1]		Exposure (years)
VehPower	Ordered discrete; mapped as categorical		Vehicle power
VehAge	Continuous (years)		Vehicle age
DrivAge	Continuous (years)		Driver age
BonusMalus	Continuous (50–350) <100 = bonus, >100 = malus		Bonus–Malus coefficient
VehBrand	Categorical		Vehicle brand
VehGas	Categorical (Diesel / Regular)		Fuel type
Area	Categories A–F		Residential density class
Density	Continuous: persons/km ²		Population density
Region	Categorical		French region
Variable	Type / Values	Description	
IDpol	Integer ID	Policy ID	
ClaimAmount	Continuous (amount)	Individual claim amount	

Variables in freMTPL2sev

freMTPL

French Motor Third-Part Liability datasets

Description

In the two datasets freMTPLfreq, freMTPLsev, risk features are collected for 413,169 motor third-part liability policies (observed mostly on one year). In addition, we have claim numbers by policy as well as the corresponding claim amounts. freMTPLfreq contains the risk features and the claim number while freMTPLsev contains the claim amount and the corresponding policy ID. Some claim amounts of freMTPLsev are fixed claim amounts based on the French IRSA-IDA claim convention, see e.g.~<https://www.index-assurance.fr/pratique/sinistre/convention-irsa>.

In the two datasets freMTPL2freq, freMTPL2sev, risk features are collected for 677,991 motor third-part liability policies (observed mostly on one year). In addition, we have claim numbers by policy as well as the corresponding claim amounts. freMTPL2freq contains the risk features and the claim number while freMTPL2sev contains the claim amount and the corresponding policy ID. Some claim amounts of freMTPL2sev are fixed claim amounts based on the French IRSA-IDA claim convention, see e.g.~<https://www.index-assurance.fr/pratique/sinistre/convention-irsa>.

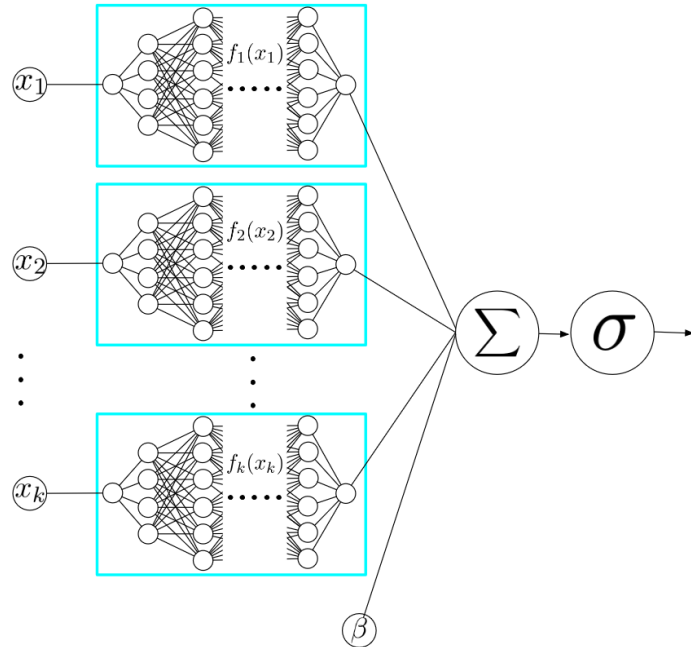
Usage

```
data(freMTPLfreq)
data(freMTPLsev)

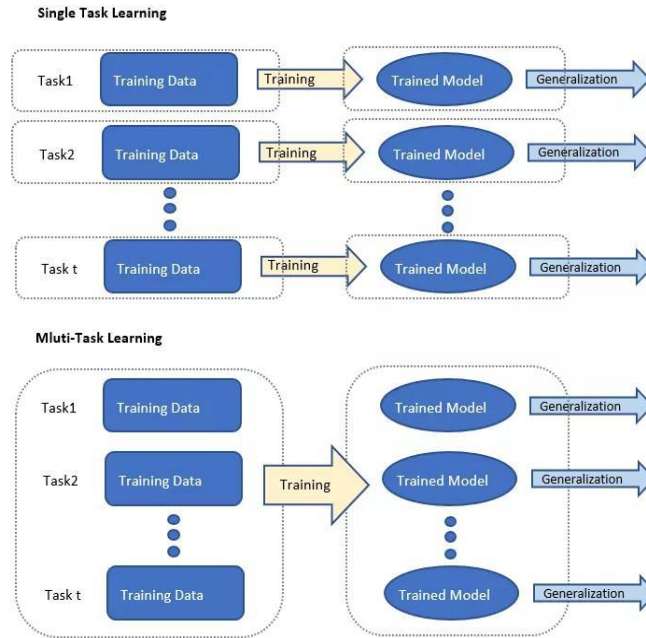
data(freMTPL2freq)
data(freMTPL2sev)
```

freMTPL is widely used in actuarial and machine-learning research. It contains policy characteristics and claim records, is publicly accessible, and is anonymized for research use.

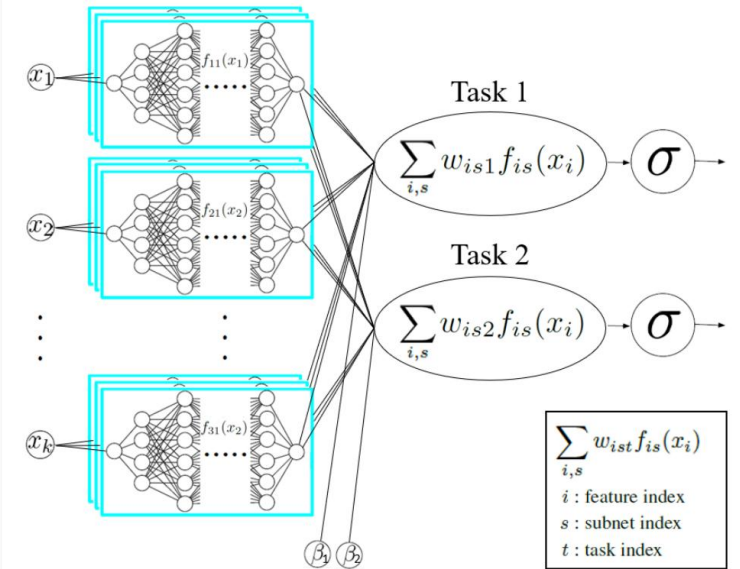
Theoretical Foundations: Neural Additive Models (NAMs) and Multi-Task Learning (MTL)



Neural Additive Model



Multi-Task Learning



Multi-Task Neural Additive Model

Core Idea

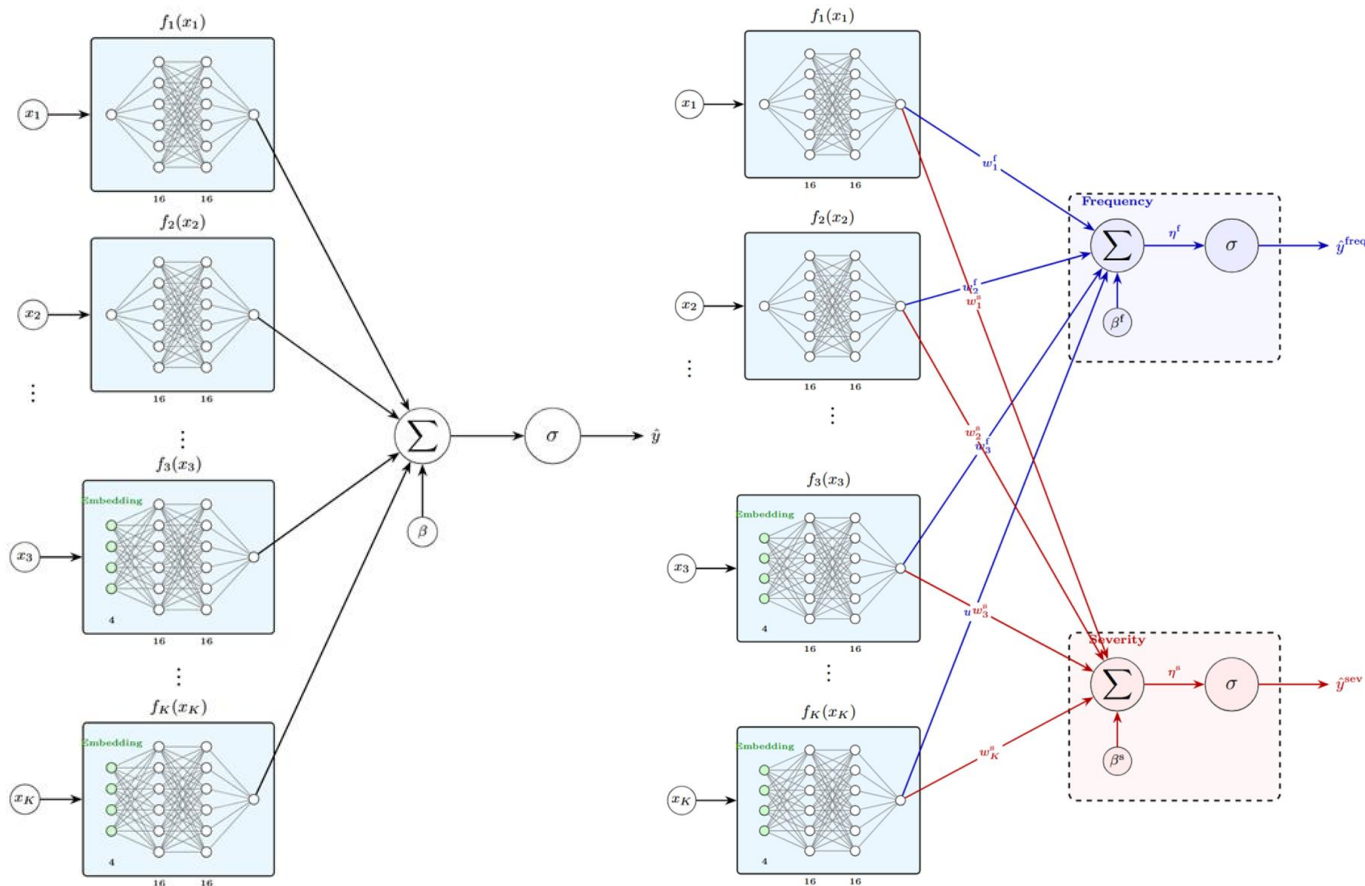
Retain an interpretable additive structure while replacing linear terms with neural subnetworks that learn nonlinear marginal feature effects.

Train related tasks jointly so they can share lower-level information, improving generalization and stability.

Treat claim frequency and severity as related tasks: share feature representations while using task-specific output layers to retain flexibility.

Model Architecture and Training

NAM (left) and MTL-NAM (right) architectures



MTL-NAM training pseudocode

算法 1 弱共享精算多任务神经加性模型 (MTL-NAM) 训练流程

Require: 训练数据 $\mathcal{D} = \{x_i, N_i, Y_i, e_i\}_{i=1}^n$; 初始网络参数 $\theta^{(0)}$; 学习率 η ; 任务损失权重 a ; 最大迭代次数 $T_{\max}$; 收敛阈值 ϵ .

Ensure: 最优模型参数 $\hat{\theta}$.

1: 初始化迭代步数 $t \leftarrow 0$

2: **repeat**

3: **Forward Pass:**

4: 对每个特征 $j = 1, \dots, p$, 计算共享形状函数:

$$f_j(x_{ij}), \quad i = 1, \dots, n$$

5: 构造频率任务线性预测子:

$$\eta_f(x_i) = \beta_f + \sum_{j=1}^p w_{f,j} f_j(x_{ij})$$

6: 构造强度任务线性预测子:

$$\eta_s(x_i) = \beta_s + \sum_{j=1}^p w_{s,j} f_j(x_{ij})$$

7: 计算双任务预测值 (log-link):

$$\hat{\lambda}_i = \exp(\eta_f(x_i)), \quad \hat{\mu}_i = \exp(\eta_s(x_i))$$

8: 计算联合加权损失:

$$\mathcal{L} = a \cdot \mathcal{L}_{\text{freq}} + (1 - a) \cdot \mathcal{L}_{\text{sev}}$$

9: **Backward Pass:**

10: 梯度下降更新参数:

$$\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \nabla_{\theta} \mathcal{L}$$

11: $t \leftarrow t + 1$

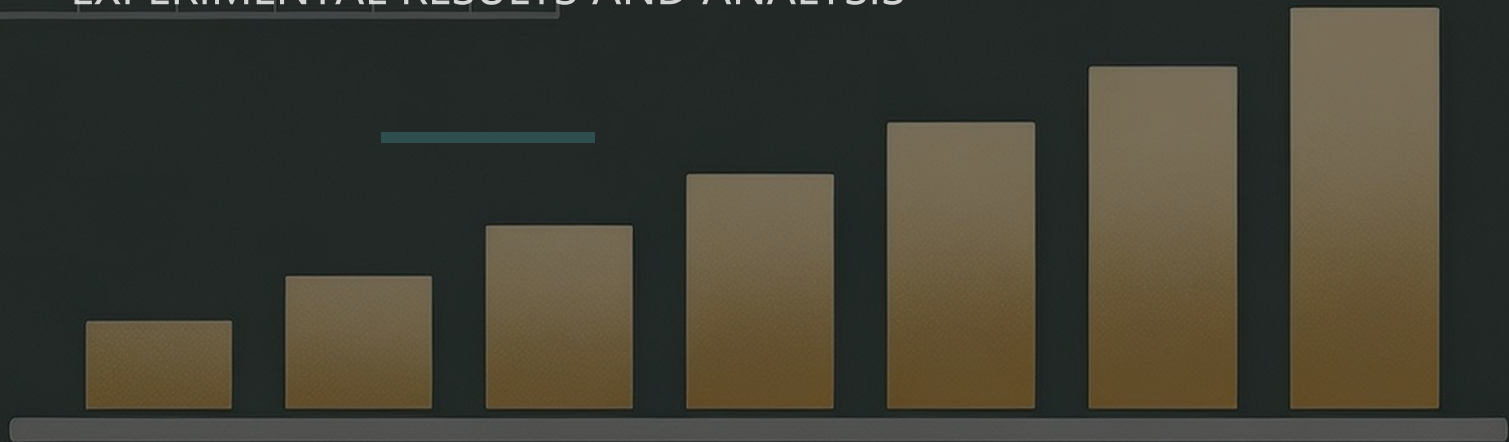
12: **until** $\|\theta^{(t+1)} - \theta^{(t)}\| < \epsilon$ or $t \geq T_{\max}$

13: **return** $\hat{\theta} = \theta^{(t+1)}$

03

Experimental Results and Analysis

EXPERIMENTAL RESULTS AND ANALYSIS



Exploratory Analysis: Distributions and Correlations

Distributional Characteristics



Claim frequency: strongly zero-inflated and heavy-tailed; most policies have no claims.



Claim severity: highly skewed and heavy-tailed, with large losses concentrated in a small number of observations.

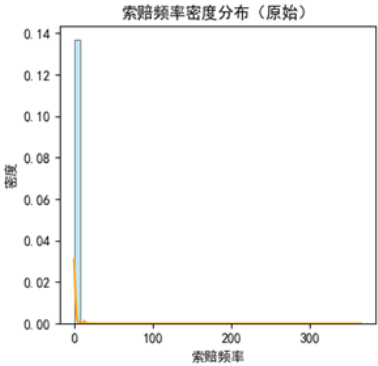
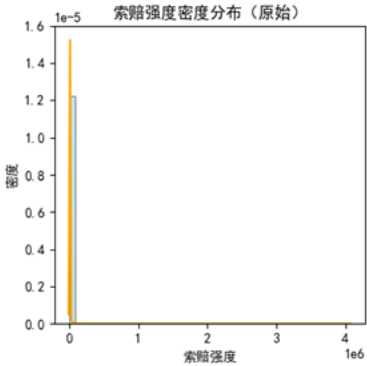
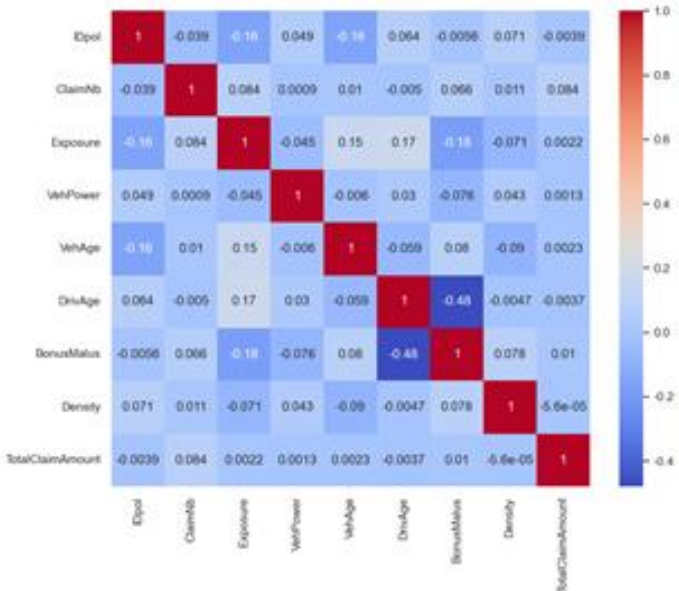
Correlation Analysis



Key negative relationship: DrivAge and BonusMalus are strongly negatively correlated, consistent with business intuition.



Weak linear correlations: most predictors have low linear correlation with the responses, suggesting substantial nonlinear structure.



Model Performance Comparison

Key Finding

Under the reported evaluation framework, MTL-NAM achieves the best values across all listed metrics and outperforms GLM and single-task NAM while remaining competitive with GBM.
The frequency–severity evaluation provides supportive evidence for MTL-NAM in pure premium modeling.

	NLL	RMSE	MAE
GLM	0.243	2.018	0.190
GBM	0.239	2.017	0.190
NAM	0.156	2.018	0.189
MTL-NAM	0.155	1.998	0.187

	NLL	RMSE	MAE
GLM	1.708	21387.021	2228.850
GBM	2.759	21369.402	1708.529
NAM	1.620	21320.620	2219.412
MTL-NAM	1.605	21003.867	1634.039

Test-set performance for claim frequency (top) and claim severity (bottom)



Best Reported Predictive Accuracy

MTL-NAM records the lowest NLL, RMSE, and MAE in the reported frequency and severity comparisons.



Multi-Task Framework Adds Value

Compared with single-task NAM, the multi-task framework improves performance by sharing information across related tasks.



Accuracy with Interpretability

The model retains inherent interpretability while achieving predictive performance competitive with a black-box GBM.

Interpretability: Shape Functions



Key Findings

MTL-NAM reveals nonlinear effects of risk factors on claim frequency and severity, with patterns broadly consistent with actuarial domain knowledge.

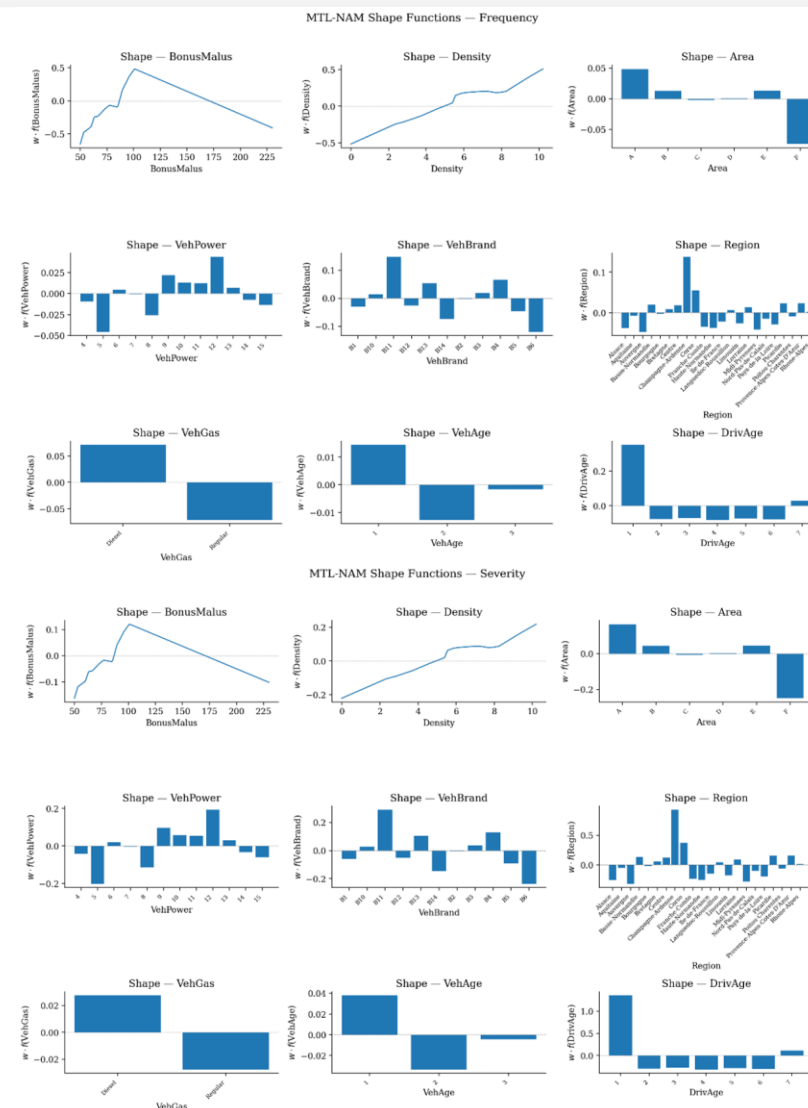
Heterogeneous Effects Across Frequency and Severity



Claim frequency: pronounced nonlinear and piecewise effects



Claim severity: different feature-effect mechanisms



MTL-NAM shape functions for claim frequency (top) and claim severity (bottom)

Interpretability: SHAP Feature Importance



Claim-Frequency Feature Importance

BonusMalus (experience-rating coefficient)

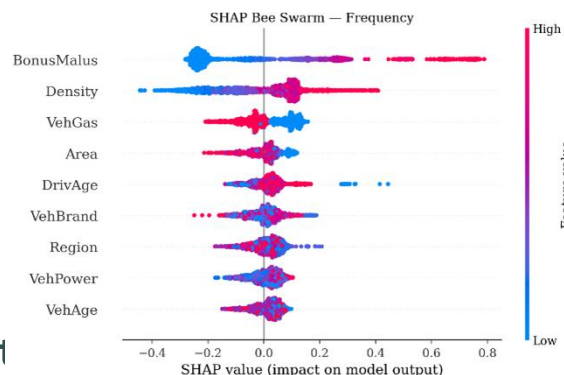
Highest-ranked risk factor for claim-frequency prediction.

VehGas (fuel type)

Secondary factor reflecting differences in claim patterns across fuel types.

Density (population density)

Key factor associated with differences in local traffic environments.



Claim-Severity Feature Importance

Region

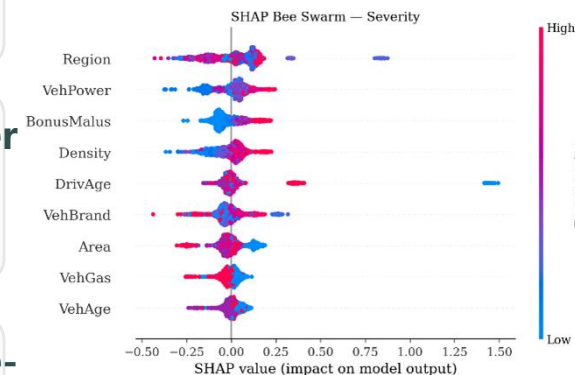
A major severity driver, potentially reflecting regional repair costs and price levels.

VehPower (vehicle power)

Higher-powered vehicles may be associated with higher repair costs and larger claim amounts.

BonusMalus (experience-rating coefficient)

Related to driving experience and claim history, with different contribution patterns across policies.



Business Interpretation

The model identifies distinct drivers of claim frequency and severity, and the patterns are broadly consistent with actuarial business intuition.



04

Conclusions, Limitations, and Future Work

CONCLUSION AND PROSPECT

Research Conclusions



Predictive Performance

Within this study's evaluation framework, MTL-NAM performs strongly, outperforming GLM and single-task NAM and remaining competitive with GBM.



Interpretability

The additive architecture provides transparent, auditable feature effects; shape functions and SHAP help quantify and interpret risk-factor contributions.



Multi-Task Learning

Joint modeling of frequency and severity allows the model to exploit shared risk information and improves generalization across the cross-validation folds.

Contributions, Limitations, and Future Work



01

Core
Contributions

Methodological Integration

Integrates multi-task learning with neural additive models for motor insurance pure premium prediction.

Unified Modeling Framework

Builds a unified framework for frequency and severity while implicitly learning shared risk structure.



02

Limitations

No Explicit Feature Interactions

The current model focuses on main effects and does not explicitly model higher-order feature interactions.

Limited Business Constraints

No explicit monotonicity constraints are imposed, so some learned effects may deviate from actuarial priors.



03

Future Work

Model Extensions

Add interaction modeling and incorporate monotonicity, smoothness, and other business-informed constraints.

Broader Applications

Explore applications in risk segmentation, fraud detection, and pricing across additional insurance lines.

Theoretical Development

Study the theoretical foundations of multi-task learning in actuarial science and identify more effective parameter-sharing structures.



Comments and suggestions are welcome

FOR LISTENING